

Muhammad Abbas Rana

+92-3221141618 | abbasrana2261@gmail.com | [LinkedIn](#) | [GitHub](#)

Computer Science student with active internship experience in ML engineering and API deployment. Working concurrently as a CS educator and online tutor, applying strong communication and analytical skills in practical, outcome-driven environments.

EDUCATION

COMSATS University (Lahore)

Bachelor of Science in Computer Science, CGPA: 3.67/4.0

Sept 2023 – Ongoing

SKILLS

Programming: Java, C++, Python; Data Structures & Algorithms; Software Development Fundamentals

Machine Learning & Data Science: Python (Pandas, NumPy, Scikit-learn), TensorFlow

Data Analysis & Visualization: SQL, Excel (Advanced Formulas, Power Query), Matplotlib, Plotly

APIs & Deployment: FastAPI, REST API Development, Docker, AWS (EC2)

Databases: MySQL, PostgreSQL, SQL Server Management Studio, Microsoft Access

Tools & Version Control: Git, GitHub

WORK EXPERIENCE

Machine Learning Engineering Intern, FlyRank AI (Remote)

Jul 2026 – Present

- Applying ML techniques including text embeddings, clustering, and intent classification to real-world search and analytics data, building toward a capstone focused on opportunity scoring.
- Producing reproducible Python notebooks using pandas and scikit-learn to extract insights and deliver actionable recommendations from production-quality datasets.

AI & Machine Learning Intern, DigiSync Technologies (Remote)

Jul 2026 – Present

- Building and structuring RESTful APIs using FastAPI, implementing Pydantic-based validation and async routing for ML-ready backend services.
- Containerizing applications with Docker and assisting with cloud deployment workflows on AWS, gaining hands-on exposure to production deployment and post-deployment monitoring practices.

Computer Science Educator, Eximus Education (Part-time, Remote)

Dec 2025 – Present

- Helped both students under examination preparation achieve top-band grades — A* at A-Level and Grade 9 at GCSE — through structured lesson delivery and targeted exam-technique coaching.
- Design lesson plans aligned to UK examination board standards, incorporating tailored exercises to address individual knowledge gaps and build exam confidence.

Online Tutor (Self-employed, Remote)

Jan 2024 – Present

- Helped all 5 Cambridge IGCSE Computer Science students achieve A* — through targeted exam preparation, past-paper practice, and one-to-one conceptual coaching.

PROJECTS

Portal Pro — Unified University Portal

Technologies: Java, JavaFX, Scene Builder, CSS, MySQL, JDBC

- Built a JavaFX desktop application with role-based portals for Admin, Teacher, and Student, supporting course, attendance, fee, salary, and marks management on a MySQL/JDBC backend.
- Designed a responsive UI in Scene Builder and implemented dynamic dashboards with role-specific user management.

Mexico Real Estate Pricing Analysis

Technologies: Python, Pandas, Matplotlib, Plotly

- Cleaned and analyzed a dataset of 21,000+ real estate listings from Properati.com using pandas, then ran exploratory and correlation analysis to identify key drivers of property prices.
- Found that property size shows only weak correlation with price in Mexico's major urban markets, concluding that location is the dominant price driver — insights visualized with Matplotlib and Plotly.

Call Centre Performance Analysis

Technologies: Excel, Power Query, Pivot Tables, Power Pivot, DAX

- Cleaned raw call centre data using Power Query and built interactive Excel dashboards with Pivot Tables, Power Pivot, and DAX to track KPIs, customer satisfaction, and sales performance.

ACHIEVEMENTS

- Completed Pakistan Youth Leadership and Dialogue Camp, Seeds of Peace (2021)
- Gold Medallist, National Robotics Competition, Punjab Group of Colleges (2022)
- 3rd Place, National Robotics Competition, Punjab Group of Colleges (2023)

VOLUNTEER EXPERIENCE

Fundraiser, The Citizens Foundation

Sept 2024 – Dec 2024

- Contributed to local fundraising as part of a team, mobilizing donations and raising awareness about education inequality through community outreach.